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Distinguishing Situations Conditions and Dependencies

effects of charge (separaton) and differentieting situations (conditioning frameworks) for fields and potentials:

Spacetime Embeding

- Spacial dependence (mechanics)
- Time dependence (dynamics)

Movement Framework

- Static: rest system, equilibrium
- Dynamic:
  - Stationary: slow motion
  - Relativistic: fast motion
  - Radiation; accelerated motion

Medium Property

- Vacuum: spacial mean = time mean
- Matter:
  - Mass (inertia)
  - Charge (conductor, ion)
  - Dielectricum (isolator)
  - Dispersion (refraction)

Field Distance

- Near Field (granular spacetime, mean average values)
- Far Field (flat spacetime)

Localisation Identity

- Global (regular)
- Local (singular)

-> Tool concepts: Momentum, Force, Energy

-> Conditions framework: - continuity condition (charge, current) - equilibrium conditions (action vs. reactio, homgenity vs. inhomogenity) - initial conditions (starting point) - border (skin) conditions (differentiation) - interior conditions (integration) - conservations of local densities (charge, energy, momentum)

-> Properties: - characteristics from charge: dielectricity ε, permeability μ, conductivity σ, frequency ω - characteristics from heat: … - Properties frm gravitation: …